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UNIVERSITI
TEKNOLOGI
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Special Topic in Computer Science (CSC649)
Big Data and Machine Learning Project

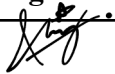
**COURIER PREFERENCES BETWEEN J&T EXPRESS AND
LALAMOVE AMONG THE PUBLIC**

11 July 2025

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Abstract

The primary objective of this study is to develop a classification model that accurately predicts courier service preference, specifically between J&T Express and Lalamove based on user demographic and service experience factors. The research focuses on analyzing key variables such as age, gender, occupation, preferred service type, delivery method, booking behavior, tracking feature usage, delivery duration, and satisfaction levels with both the delivery process and personnel. By applying statistical analysis and machine learning techniques to a dataset of 500 survey entries, this study aims to identify significant patterns and correlations that influence courier preferences. The integration of public survey data provides valuable insights into customer behavior, helping logistics companies optimize service quality, improve resource allocation, and better align offerings with user expectations.

Keywords: courier preference, delivery satisfaction, tracking feature, machine learning, J&T Express, Lalamove, public survey analysis

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List of Abbreviations

ABBREVIATION	MEANING
ANN	Artificial Neural Network
GPS	Global Positioning System
IoT	Internet of Things
KNN	K-Nearest Neighbors
K-Fold	K-Fold Cross Validation (a machine learning validation technique)
ML	Machine Learning
SEM	Structural Equation Modeling
UiTM	Universiti Teknologi MARA

1.0 INTRODUCTION

As courier services become a bigger part of everyday life in Malaysia, especially with the rise of online shopping and small business deliveries, many people find themselves relying on platforms like J&T Express and Lalamove. These two services are among the most commonly used, each offering different strengths that cater to different needs. However, despite their popularity, there's still a lot of uncertainty about which service works best in various situations.



Figure 1.0: J&T Headquarters

(Source: <https://www.ceriasihat.com/jt-perkhidmatan-kurier-terbaik-di-malaysia/>, n.d)

This study was carried out to better understand how people actually feel about using J&T Express and Lalamove. We wanted to know what influences their choices, what their experiences have been like, and which service they find more reliable or convenient. To do this, we created and distributed a survey to a wide range of users, including students, small business owners, and everyday consumers.

The goal was to gather real opinions from real users. By listening to their feedback, we hope to get a clearer picture of what people expect from courier services, what matters most to them is whether it's speed, cost, or customer support and which provider meets those expectations more effectively. This study was designed not only to help users make more informed choices but also to give courier companies insights on how they can improve and serve their customers better.

2.0 PROBLEM STATEMENT

In Malaysia's rapidly evolving e-commerce and delivery ecosystem, courier services play a vital role in connecting businesses to customers efficiently. Among the many providers, J&T Express and Lalamove have become two of the most commonly used options, each catering to different logistics needs. J&T Express is favored for its nationwide reach and affordability, while Lalamove is known for its fast, real-time delivery within urban areas. Despite their growing popularity, there remains a lack of clear understanding about which courier service is most preferred by users, and what factors influence that choice. These problems often lead to confusion and dissatisfaction, especially among small businesses and individuals who rely heavily on courier services for daily operations or personal needs.

Both couriers present distinct advantages and disadvantages that appeal to different user segments. J&T Express offers competitive rates and parcel tracking, making it ideal for online sellers and long-distance deliveries. However, it is frequently criticized for delivery delays, parcel mishandling, and poor customer support. Lalamove, on the other hand, stands out with its same-day delivery, GPS tracking, and vehicle flexibility. Unfortunately this courier is often reported to be expensive for longer trips and unavailable in rural areas. These differences highlight the trade-offs users must consider when selecting a delivery service, especially when balancing speed, cost, and reliability.

To have a better understanding of a public preference and identify which courier service better meets user expectations, our team conducted a survey using Google Forms, targeting a broad range of respondents including students, online sellers, and everyday users. This form was distributed to a wide audience, including students, small business operators, and everyday users with delivery experience. The survey focused on gathering real-life opinions and user experiences related to pricing, delivery speed, service reliability, convenience, and overall satisfaction. Collecting responses from a diverse demographic helped ensure that the feedback reflects a broad and accurate view of public expectations.

After gathering the data, we analyzed the responses to identify trends, preferences, and pain points associated with both services. This analysis aims to determine which courier service is generally preferred by the public and why, offering valuable insights for both service providers and consumers. The findings are expected to guide individuals and businesses in making more informed decisions while also highlighting areas where each courier can improve. In the end, this analysis aims to assist in recommending the most suitable courier service depending on specific user needs and usage contexts, and could serve as the basis for a decision-making tool or logistics improvement strategy.

3.0 PROJECT OBJECTIVE

- To evaluate user preferences between J&T Express and Lalamove in Malaysia in delivering the parcel from one location to another .
- To identify key factors that influencing courier service choice, such as:
 - Package handling
 - Delivery booking platform
 - Service reliability
 - Delivery type
 - Customer service
- To compare the overall user satisfaction levels for both courier services and determine which courier service better meets public expectations.

4.0 PROJECT SCOPE

This project focuses on understanding user preferences and experiences with two popular courier services in Malaysia which is J&T Express and Lalamove and using a survey-based approach to collect and analyze data through data analytics techniques. The core objective is not only to gather public opinion but also to process this data to uncover patterns, trends, and key insights through structured analysis.

The scope of the study involves designing and distributing a digital survey using Google Forms. The survey targeted a diverse group of respondents, including students, online sellers, and general consumers with prior experience using J&T Express, Lalamove, or both. The survey collected both quantitative and qualitative data across several key areas, such as delivery speed, reliability, platform usability, customer service quality, and overall satisfaction.

Once collected, the data was cleaned, organized, and prepared for analysis. It was then split into training and testing datasets, allowing us to apply data learning techniques and predictive analytics. This process enabled the team to:

- **Train models** using real user data to identify the most influential factors in courier service selection.
- **Test predictions** and validate assumptions based on user preferences and satisfaction.
- **Analyze correlations** between user demographics and their expectations or experiences.
- **Generate results** that provide a clearer picture of which courier service is preferred and why.

This data-driven approach expands the scope beyond simple feedback collection as it transforms the survey into a foundation for deeper learning and meaningful decision-making. By applying basic data analytics tools and techniques, the study aims to deliver insights that are statistically supported and actionable.

The project focuses solely on domestic courier services within Malaysia and is limited to user-supplied data. It does not include operational or technical data from the courier companies themselves. The analysis is based entirely on user perception and experience, making the findings relevant for consumers, small businesses, and service providers seeking to improve their performance.

In summary, the scope of this project includes survey design, data collection, data preparation, and analysis using basic machine learning principles such as training and testing sets. The goal is to provide a data-informed understanding of public preferences between J&T Express and Lalamove, helping users make better choices and enabling service providers to identify areas for improvement.

5.0 SIGNIFICANCE OF PROJECT

This project holds significance in several important ways, particularly in the context of Malaysia's growing reliance on courier services for both personal and business purposes. As e-commerce continues to expand and consumers increasingly depend on fast, reliable delivery, understanding which courier service meets user needs more effectively has become crucial.

By collecting real-world data through surveys and applying data analytics techniques, this project offers valuable insights that go beyond surface-level opinions. It uses data as a tool to identify patterns, preferences, and problem areas in the user experience of two major courier services, J&T Express and Lalamove. These insights are meaningful not only for consumers who are trying to make informed choices, but also for service providers looking to enhance their offerings.

In a consumer perspective, the project helps individuals and businesses better understand the strengths and weaknesses of each courier service. This supports smarter decision-making based on specific needs such as delivery speed, pricing, convenience, and customer service. For small businesses, in particular, choosing the right courier can directly impact customer satisfaction and operational efficiency.

From a logistics and service provider perspective, the findings highlight user expectations and recurring issues. This feedback can be used to improve delivery processes, app interfaces, customer support systems, and overall service quality. Companies can also use the data to refine their marketing strategies by focusing on what matters most to users.

In addition, the project demonstrates the practical use of data analytics in solving real-world problems. By treating survey responses as a dataset for training and testing, the project introduces basic data science concepts in a meaningful way. This adds academic and practical value, especially for students or researchers interested in applying analytical skills to social or consumer behavior studies.

Ultimately, this project contributes to a better understanding of Malaysia's courier service landscape by bridging the gap between user experience and service performance. It provides evidence-based recommendations that can drive improvements in delivery services and empower users with knowledge to make confident, informed choices.

6.0 LITERATURE REVIEW

Recent advancements in data science have led to increased interest in using survey data not just for descriptive analysis but also for training machine learning models to uncover patterns, predict behavior, and support decision-making. Traditional survey studies, while valuable for collecting user opinions, are now being combined with supervised learning techniques to gain deeper insights. In this context, survey data serves as a valuable dataset for classification, clustering, and regression models. Researchers have shown that properly structured and cleaned survey responses can be split into training and testing datasets, allowing for the creation of models that identify key factors driving user preferences, satisfaction, and service selection.

For instance, Ong et al. (2023) applied classification models to survey data to understand workplace preferences among university graduates. The study used common machine learning practices such as splitting the dataset into training and test sets, followed by the application of decision trees, logistic regression, and support vector machines to predict outcomes with high accuracy. This research highlights how structured opinion data, originally collected for feedback so it can be repurposed for model training, allows for not just understanding but prediction of user behavior. Similarly, Ahmed et al. (2020) combined survey-based Structural Equation Modeling (SEM) with Artificial Neural Networks (ANN) to forecast the adoption of IoT services in smart mobility contexts. Their hybrid approach trained models using survey inputs and validated them with testing data, achieving predictive capabilities that surpassed traditional analysis methods.

These studies illustrate that survey data is no longer limited to reporting trends but can function as a training foundation for intelligent systems, enabling predictive modeling and recommendation systems. In the context of this project, where user preferences for courier services such as J&T Express and Lalamove are collected through surveys, this data can be used to train models that predict user satisfaction based on features like package handling, booking convenience, delivery type, and customer service. By splitting the dataset into training and testing subsets, the project adopts a practical approach similar to those in previous research, using real-world responses to simulate and validate decision-making scenarios.

This shift from simple feedback gathering to data-driven learning demonstrates the evolving role of survey data in analytics. With proper preprocessing and feature selection, such datasets can support not only evaluation but also model accuracy testing and decision support. As Malaysia's courier and logistics industry grows more competitive, understanding and anticipating user needs through data learning methods provides a distinct strategic advantage, both for service providers and researchers. This project contributes to that growing field by using survey results not just for summary insights, but as a functional dataset to develop, train, and test models that reflect real consumer behavior in the courier industry.

7.0 DATA COLLECTION

The data collection process for this project was carried out using an online survey developed through Google Forms. The survey was shared across various digital platforms to ensure it reached a broad and diverse audience. Respondents included students, working professionals, freelancers, online sellers, and small business owners, the groups that commonly use courier services for either personal or business-related deliveries. The survey was designed to be straightforward and accessible, allowing participants to provide honest feedback quickly and easily, regardless of their background or technical ability.

Collecting data directly from users was essential for capturing real-world experiences with courier services, particularly J&T Express and Lalamove. These firsthand accounts offered valuable insight into users' expectations, frustrations, and preferences, forming a reliable basis for understanding the strengths and weaknesses of each service. By gathering data from individuals with varying delivery needs and habits, the study ensured that the results would reflect a wide range of practical use cases, making the findings more comprehensive and applicable.

The importance of this data collection lies in its ability to bridge the gap between service performance and user perception. In an industry where reliability and user satisfaction are critical, understanding how people actually experience these services is crucial. The data provided a structured way to quantify user sentiment and observe emerging trends, enabling an evidence-based approach to evaluating courier performance. Without this direct input from users, it would be difficult to draw meaningful conclusions or suggest improvements grounded in actual user behavior.

	A	B	C	D	E	F	G	H	I	J	K
1	Age	Gender	Occupation	Preferred_Service	Types_Delivery	Book_Delivery	Tracking_Feature	Delivery_Days	Satisfaction_of_Delivery	Satisfaction_of_Deliver_Person	Courier_Preference
2	45	Female	Freelancer	J&T Express	Express	Walk-in	Yes	1	3		3 J&T Express
3	22	Female	Student	J&T Express	Express	Mobile App	Yes	3	4		3 J&T Express
4	43	Female	Office Worker	J&T Express	Scheduled delivery	Mobile App	Yes	4	4		5 J&T Express
5	24	Female	Office Worker	J&T Express	Express	Walk-in	Yes	4	3		4 J&T Express
6	43	Male	Office Worker	Neither	Economy (low-cost)	Website	No	1	3		3 J&T Express
7	22	Female	Student	J&T Express	Next-day	Walk-in	Yes	3	4		3 J&T Express
8	52	Female	Office Worker	J&T Express	Scheduled delivery	Mobile App	Yes	3	4		4 J&T Express
9	22	Female	Small Business Owner	J&T Express	Scheduled delivery	Mobile App	No	3	4		4 J&T Express
10	34	Female	Freelancer	J&T Express	Next-day	Mobile App	Yes	3	5		5 J&T Express
11	19	Male	Student	J&T Express	Scheduled delivery	Mobile App	Yes	5	4		5 J&T Express
12	20	Male	Student	J&T Express	Express	Mobile App	No	1	3		3 J&T Express
13	20	Male	Student	J&T Express	Express	Mobile App	No	3	3		3 J&T Express
14	41	Male	Office Worker	J&T Express	Express	Mobile App	Yes	2	5		5 J&T Express
15	22	Male	Student	J&T Express	Express	Mobile App	Yes	5	5		5 J&T Express
16	47	Female	Office Worker	J&T Express	Next-day	Walk-in	Yes	2	4		5 J&T Express
17	49	Female	Small Business Owner	J&T Express	Express	Mobile App	Yes	5	4		4 J&T Express
18	49	Female	Small Business Owner	Lalamove	Same-day	Mobile App	Yes	1	4		4 Lalamove
19	22	Female	Student	J&T Express	Express	Mobile App	Yes	3	4		4 J&T Express
20	47	Female	Office Worker	J&T Express	Scheduled delivery	Mobile App	No	2	4		4 J&T Express
21	49	Female	Office Worker	J&T Express	Express	Mobile App	Yes	3	4		4 J&T Express
22	49	Female	Office Worker	J&T Express	Express	Website	Yes	4	5		5 J&T Express
23	46	Female	Office Worker	J&T Express	Scheduled delivery	Mobile App	Yes	3	4		4 J&T Express
24	49	Female	Office Worker	J&T Express	Scheduled delivery	Mobile App	Yes	2	2		4 J&T Express

Figure 7.0 Dataset Collection

7.1 Data Analytic Purposes

The data collected from the survey was prepared and structured for analytical purposes, particularly to support machine learning tasks such as classification and pattern detection. The dataset was cleaned and organized to ensure consistency and accuracy before being divided into training and testing sets. This allowed for model development and evaluation, providing the opportunity to make predictions based on user preferences and service experiences.

By analyzing both categorical and numerical data, the study aimed to uncover deeper insights into what influences courier service preferences and satisfaction. The categorical data helped identify common trends and group behaviors, while the numerical data supported the assessment of service quality and user satisfaction through measurable metrics. Together, these data types formed a comprehensive dataset suitable for supervised learning models and statistical analysis.

7.1.1 Categorical Data

The categorical data in our dataset are gender, occupation, preferred service, types of delivery chosen, how the delivery is booked, and tracking feature.

```
categorical_data = dfCSV.select_dtypes(include=['object']).columns
categorical_data = [col for col in categorical_data if col not in ['Timestamp', 'Courier_Preference']]
dfCSV[categorical_data].info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 501 entries, 0 to 500
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Gender                 501 non-null   object
1   Occupation              501 non-null   object
2   Preferred_Service       501 non-null   object
3   Types_Delivery         501 non-null   object
4   Book_Delivery           501 non-null   object
5   Tracking_Feature       501 non-null   object
dtypes: object(6)
memory usage: 23.6+ KB
```

Figure 7.1 Categorical Data

7.1.2 Numerical Data

The numerical data in our dataset are age, delivery days, satisfaction of delivery, and satisfaction towards the person that delivers their package.

```
numerical_data = dfCSV.select_dtypes(include=['int64', 'float64']).columns
dfCSV[numerical_data].info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 501 entries, 0 to 500
Data columns (total 4 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   501 non-null    int64
1   Delivery_Days                        501 non-null    float64
2   Satisfaction_of_Delivery             501 non-null    int64
3   Satisfaction_of_Deliver_Person       501 non-null    int64
dtypes: float64(1), int64(3)
memory usage: 15.8 KB
```

Figure 7.2 Numerical Data

7.2 Data Science Purposes

7.2.1 Label

The label refers to the target variable for the outcome that the model wants to predict. It stands for the particular dataset feature that is the subject of analysis or prediction. In this case, “courier preference” is the label for this dataset, which contains both J&T Express and Lalamove.

```
result_counts = dfCSV['Courier_Preference'].value_counts()
print(result_counts)

Courier_Preference
J&T Express      274
Lalamove         227
Name: count, dtype: int64
```

Figure 7.3 Label Target

7.2.2 Features

We have collected data from our respondents based on their courier preferences. The features included in the figure below.

	Age	Gender	Occupation	Prefered_Service	Types_Delivery	Book_Delivery	Tracking_Feature	Delivery_Days	Satisfaction_of_Delivery	Satisfaction_of_Deliver_Person
0	45	0	0	1	1	2	1	1.0	3	3
1	22	0	4	1	1	0	1	3.0	4	3
2	43	0	1	1	4	0	1	4.0	4	5
3	24	0	1	1	1	2	1	4.0	3	4
4	43	1	1	3	0	3	0	1.0	3	3

Figure 7.5 List of Features

7.3 Target Respondent

7.3.1 Data Collection Instrument (e.g., questionnaire)

Table 7.0 Data Collection Instrument

Section	Description of Selection	Question	Type of Variables
1	What is your age?	1	Numerical
	What is your gender?	2	Categorical
	What is your occupation?	3	Categorical
	Which service have you used more frequently?	4	Categorical
	What type of delivery do you usually choose?	5	Categorical
	How do you usually book your delivery?	6	Categorical
	Do you use the tracking feature	7	Categorical
	How many days on average, did your delivery take?	8	Numerical
	How satisfied were you with the handling of your package during delivery?	9	Numerical
	How satisfied were you with the professionalism of the delivery personnel?	10	Numerical
	Which courier service do you prefer overall?	11	Target

7.3.2 Distribution Strategy

The distribution strategy for the survey is through email and social media. Email is used to reach UiTM students in UiTM Jasin in Melaka. Social media such as Telegram, Instagram and WhatsApp. Both methods allow us to obtain various results from the public.

8.0 Data Science

8.1 Data Preprocessing

8.1.1 Data Cleaning

Data cleaning to ensure no null values in the dataset. This is important because the survey often contains noise and missing value. To resolve this issue, data cleaning is conducted to address this issue. The figure below shows the data cleaning process in this model resulting in the dataset without any null values.

Table 8.0 Output after data cleaning

Source Code	Output
<pre>#To report missing data dfCSV.apply(lambda x: sum(x.isnull()), axis=0)</pre>	<pre>Age 0 Gender 0 Occupation 0 Preferred_Service 0 Types_Delivery 0 Book_Delivery 0 Tracking_Feature 0 Delivery_Days 0 Satisfaction_of_Delivery 0 Satisfaction_of_Deliver_Person 0 Courier_Preference 0 dtype: int64</pre>

8.1.2 Data Selection (Data Slicing for x = features, y = label)

At the data selection stage, the data are divided into two parts, “x” represents the features attributes while “y” represents the label which indicates the type.

Table 8.1 Output after data cleaning

Source Code	Output
<pre>#Data slicing #x = dfCSV.iloc[:, [1,3,4,5,6,7,8,9,10,11,12]] x = dfCSV.drop(['Courier_Preference'], axis = 1) y = dfCSV.iloc[:, [10]] x.head()</pre>	<pre>Age Gender Occupation Preferred_Service Types_Delivery Book_Delivery Tracking_Feature Delivery_Days Satisfaction_of_Delivery Satisfaction_of_Deliver_Person 0 45 0 0 1 1 2 1 1.0 3 3 1 22 0 4 1 1 0 1 3.0 4 3 2 43 0 1 1 4 0 1 4.0 4 5 3 24 0 1 1 1 2 1 4.0 3 4 4 43 1 1 3 0 3 0 1.0 3 3</pre>

8.1.3 Data Transformation

Different data types are allowed in the dataset, some are numerical while others can be categorical. The categorical can be defined as a data type that indicates a reason. The Label Encoder function lets you change the category data to numeric data. Table 8.3 shows the coding of transformed data.

Table 8.2 Output after data cleaning

Source Code	Output																																																																		
<pre>from sklearn.preprocessing import LabelEncoder var_mod = ['Gender', 'Occupation', 'Preferred_Service', 'Types_Delivery', 'Book_Delivery', 'Tracking_Feature', 'Courier_Preference'] le = LabelEncoder() for i in var_mod: dfCSV[i] = le.fit_transform(dfCSV[i]) dfCSV.head()</pre>	<table><tr><th>Age</th><th>Gender</th><th>Occupation</th><th>Preferred_Service</th><th>Types_Delivery</th><th>Book_Delivery</th><th>Tracking_Feature</th><th>Delivery_Days</th><th>Satisfaction_of_Delivery</th><th>Satisfaction_of_Delivery_Person</th><th>Courier_Preference</th></tr><tr><td>40</td><td>0</td><td>0</td><td>1</td><td>1</td><td>2</td><td>1</td><td>1.0</td><td>3</td><td>3</td><td>0</td></tr><tr><td>22</td><td>0</td><td>4</td><td>1</td><td>1</td><td>0</td><td>1</td><td>3.0</td><td>4</td><td>3</td><td>0</td></tr><tr><td>40</td><td>0</td><td>1</td><td>1</td><td>4</td><td>0</td><td>1</td><td>4.0</td><td>4</td><td>3</td><td>0</td></tr><tr><td>24</td><td>0</td><td>1</td><td>1</td><td>1</td><td>2</td><td>1</td><td>4.0</td><td>3</td><td>4</td><td>0</td></tr><tr><td>40</td><td>1</td><td>1</td><td>3</td><td>0</td><td>3</td><td>0</td><td>1.0</td><td>3</td><td>3</td><td>0</td></tr></table>	Age	Gender	Occupation	Preferred_Service	Types_Delivery	Book_Delivery	Tracking_Feature	Delivery_Days	Satisfaction_of_Delivery	Satisfaction_of_Delivery_Person	Courier_Preference	40	0	0	1	1	2	1	1.0	3	3	0	22	0	4	1	1	0	1	3.0	4	3	0	40	0	1	1	4	0	1	4.0	4	3	0	24	0	1	1	1	2	1	4.0	3	4	0	40	1	1	3	0	3	0	1.0	3	3	0
Age	Gender	Occupation	Preferred_Service	Types_Delivery	Book_Delivery	Tracking_Feature	Delivery_Days	Satisfaction_of_Delivery	Satisfaction_of_Delivery_Person	Courier_Preference																																																									
40	0	0	1	1	2	1	1.0	3	3	0																																																									
22	0	4	1	1	0	1	3.0	4	3	0																																																									
40	0	1	1	4	0	1	4.0	4	3	0																																																									
24	0	1	1	1	2	1	4.0	3	4	0																																																									
40	1	1	3	0	3	0	1.0	3	3	0																																																									

8.2 Classification Model (Machine learning approach on big data project)

8.2.1 Data Split Approach

There are 501 total data in the dataset, which will be divided into 80:20 ratios. In this approach, 80% is utilized for training and 20% is used for testing. The K-Fold approach is used, which divides the dataset into K-Folds, or equal-sized subsets.

1. Divide the dataset into 5 equal subsets $501/5 = 100.1$.
2. Shuffle the dataset to get accurate data that matches the target.
3. Train the machine learning model on 4 of these parts and validate it on the remaining part.
4. Repeat the process for all parts, and rotate which part is used for validation each time.
5. Evaluate the model's performance across all parts and average the results the overall

effectiveness and reliability.

8.2.2 Modelling Classification Model

8.2.2.1 Machine Learning Algorithm (E.g., KNN)

One of the primary methods in machine learning for classification is K-Nearest Neighbours. It is heavily utilized to identify patterns, mines, instructions and is a component of the supervised learning domain. Due to its non-parametric nature, which eliminates the need for underlying assumptions about data distribution, it is frequently used in real-world situations. As a result, we are calculating the distance between the two points in the plane with coordinates (x, y) and (a, b) using Euclidean distance to determine who our nearest neighbours are. This formula is displayed in Figure 8.1 below.

```
# K-Nearest Neighbor
from sklearn.neighbors import KNeighborsClassifier
ModelKITA = KNeighborsClassifier()

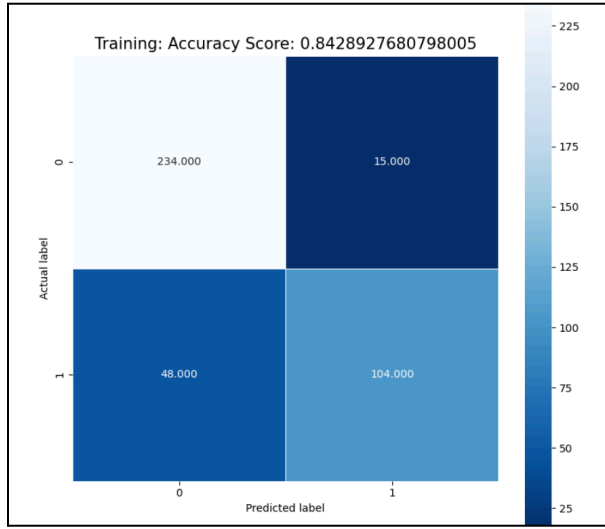
ModelKITA.fit(x, y)
yPred = ModelKITA.predict(x)
```

Figure 8.0 K-Nearest Neighbor Source Code

8.2.2.2 Testing and Performance Analysis

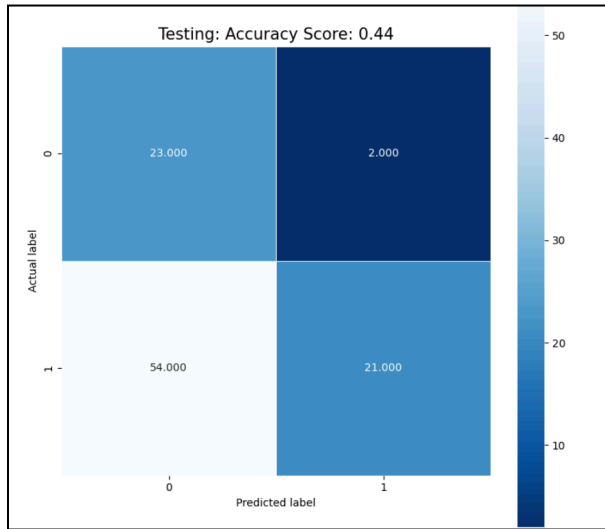
Data Train Assessment

Table 8.3 Data Train Default

Source Code	Output									
<pre># Testing & performance analysis - Training Data import sklearn.metrics as skm import matplotlib.pyplot as plt import seaborn as sns plt.figure(figsize=(9,9)) sns.heatmap(skm.confusion_matrix(y, yPred), annot=True, fmt=".3f", linewidths=.5, square = True, cmap = 'Blues_r'); plt.ylabel('Actual label'); plt.xlabel('Predicted label'); all_sample_title = 'Training: Accuracy Score: {0}'.format(skm.accuracy_score(y, yPred)) plt.title(all_sample_title, size = 15);</pre>	 Training: Accuracy Score: 0.8428927680798005 <table><tr><th>Actual \ Predicted</th><th>0</th><th>1</th></tr><tr><th>0</th><td>234.000</td><td>15.000</td></tr><tr><th>1</th><td>48.000</td><td>104.000</td></tr></table>	Actual \ Predicted	0	1	0	234.000	15.000	1	48.000	104.000
Actual \ Predicted	0	1								
0	234.000	15.000								
1	48.000	104.000								

Data Test Assessment

Table 8.4 Test Default

Source Code	Output									
<pre># Testing & performance analysis -Testing data import sklearn.metrics as skm import matplotlib.pyplot as plt import seaborn as sns plt.figure(figsize=(9,9)) sns.heatmap(skm.confusion_matrix(y2, yPred2), annot=True, fmt=".3f", linewidths=.5, square = True, cmap = 'Blues_r'); plt.ylabel('Actual label'); plt.xlabel('Predicted label'); all_sample_title = 'Testing: Accuracy Score: {0}'.format(skm.accuracy_score(y2, yPred2)) plt.title(all_sample_title, size = 15);</pre>	 Testing: Accuracy Score: 0.44 <table><tr><th>Actual label \ Predicted label</th><th>0</th><th>1</th></tr><tr><th>0</th><td>23.000</td><td>2.000</td></tr><tr><th>1</th><td>54.000</td><td>21.000</td></tr></table>	Actual label \ Predicted label	0	1	0	23.000	2.000	1	54.000	21.000
Actual label \ Predicted label	0	1								
0	23.000	2.000								
1	54.000	21.000								

8.2.3 Cross-validation Approach (K-Fold)

Cross-validation k-fold has many benefits that can be used advantageously. Compared to a single train-test split, it offers a more tried and tested evaluation of the model's performance. By using several folds reduce the similarity of biases and variances that may rise because of specific data split. Secondly, the cross-validation k-fold algorithm uses the data provided efficiently by making use of each data point for both training and testing. This leads to a more comprehensive evaluation of the model. In conclusion, k-fold cross-validation can help with hyperparameter optimization. K-fold cross-validation entails iterating k times over the training set with different hyperparameter values and finding those that yield the highest performance within the k fold process.

8.2.3.1 List of K-Fold Experiments Batch

This is the best top five K-Fold experiments from 50 batches.

Experiment batch 2: Experiment P=1, n_neighbors=2

Accuracy Training = 91.3

Accuracy Testing = 84

#EXPERIMENT P =1, n_neighbors = 2

	TRAIN	TEST
KF1	88.7	44
KF2	91.5	13
KF3	90.3	91
KF4	91.3	84
KF5	88.8	44
AVERAGE	90.12	55.2

Figure 8.1 K-Fold Batch 2

Experiment batch 4: Experiment P=1, n_neighbors=5

Accuracy Training = 89.3

Accuracy Testing = 88

#EXPERIMENT P =1, n_neighbors = 4

	TRAIN	TEST
KF1	86.8	44
KF2	89.5	15
KF3	89.3	88
KF4	89	83
KF5	86.8	44
AVERAGE	88.28	54.8

Figure 8.2 K-Fold Batch 4

Experiment batch 2: Experiment P=1, n_neighbors=2

Accuracy Training = 88

Accuracy Testing = 88

#EXPERIMENT P =1, n_neighbors = 6

	TRAIN	TEST
KF1	83.8	37
KF2	88.5	15
KF3	88	88
KF4	87.8	82
KF5	83.8	37
AVERAGE	86.38	51.8

Figure 8.3 K-Fold Batch 2

Experiment batch 5: Experiment P=1, n_neighbors=5

Accuracy Training = 89.3

Accuracy Testing = 85

#EXPERIMENT P =1, n_neighbors = 5

	TRAIN	TEST
KF1	87.5	57
KF2	90.7	22
KF3	89.3	85
KF4	90.8	79
KF5	87.5	57
AVERAGE	89.16	60

Figure 8.4 K-Fold Batch 5

Experiment batch 22: Experiment P=2, n_neighbors=2

Accuracy Training = 88.8

Accuracy Testing = 86

#EXPERIMENT P =2, n_neighbors = 2

	TRAIN	TEST
KF1	87.3	38
KF2	90.5	14
KF3	88.8	86
KF4	89	83
KF5	87.3	38
AVERAGE	88.58	51.8

Figure 8.5 K-Fold Batch 22

8.2.3.2 Proposed the Best K-Fold Experiment Batch (The best hyperparameter combination)

Figure 8.2 showcases the results of the best K-Fold experiment from the model, displaying performance measures and outcomes. Specifically, K-Fold 4 demonstrates the highest testing accuracy compared to other experiments.

#EXPERIMENT P =1, n_neighbors = 2

	TRAIN	TEST
KF1	88.7	44
KF2	91.5	13
KF3	90.3	91
KF4	91.3	84
KF5	88.8	44
AVERAGE	90.12	55.2

Figure 8.6 Result of the Best K-Fold Experiment Batch

8.2.4 Remodelling Classification Model (Applying K-Fold Result)

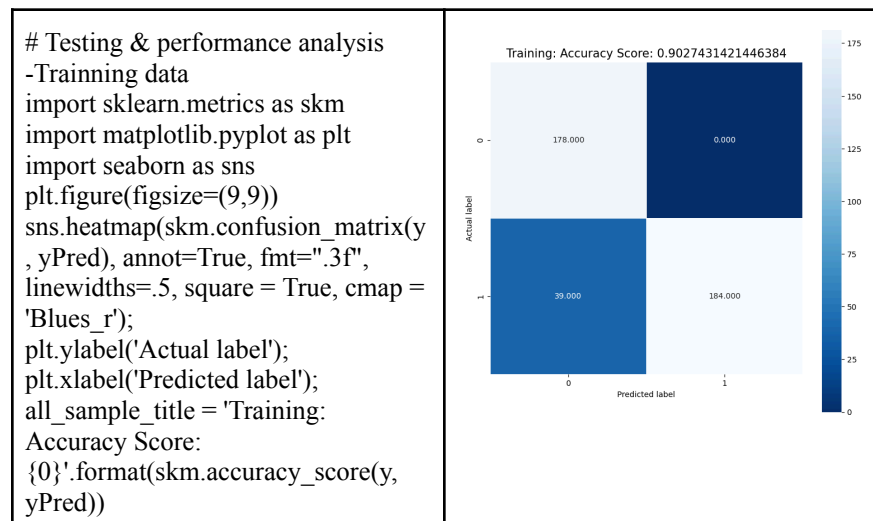
Using the knowledge from the best -performing K-Fold trial, the classification model is being redesigned to improve upon the current model. By optimizing the model's design and parameters inactively, classification performance and accuracy are improved.

8.2.4.1 Testing and Performance Analysis

The model was run on default parameters and the result of the training accuracy score was 90.3% percent and the testing accuracy score was 91% percent.

Data Train Assessment

Table 8.5 Code and Result of Accuracy Score for Training

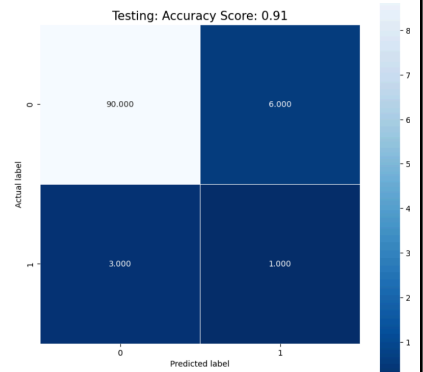


```
plt.title(all_sample_title, size = 15);
```

Data Test Assessment

Table 8.6 Code and Result of Accuracy Score for Testing

```
# Testing & performance analysis
-Testing data
import sklearn.metrics as skm
import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(9,9))
sns.heatmap(skm.confusion_matrix(y
2, yPred2), annot=True, fmt=".3f",
linewidths=.5, square = True, cmap =
'Blues_r');
plt.ylabel('Actual label');
plt.xlabel('Predicted label');
all_sample_title = 'Testing: Accuracy
Score:
{0}'.format(skm.accuracy_score(y2,
yPred2))
plt.title(all_sample_title, size = 15);
```

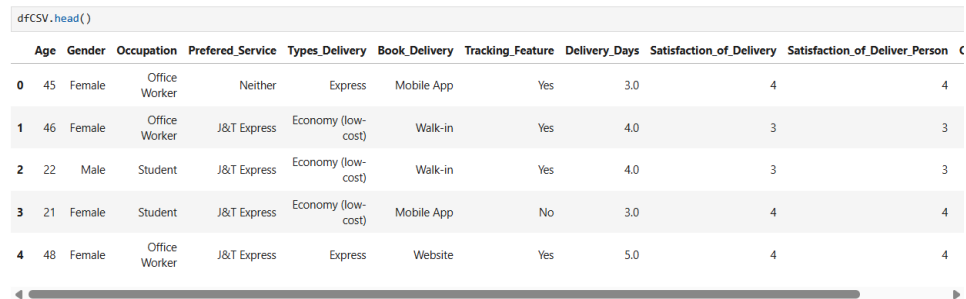


8.3 Product (Prototype of application for prediction project)

In addition to the aims already stated, this research paper will examine how specific service-related factors influence users' choices between courier services. Our goal is to understand how user comfort and satisfaction are affected by the usefulness and convenience of the courier service, particularly in terms of delivery efficiency, tracking features, and overall service experience.

8.3.1 Applying the Classification Model in Application

Utilizing the categorization model in the application functions as a validation process to verify the accuracy of coding. Firstly, evaluate the model's performance by utilizing the data collected from respondents to confirm if the generated outputs are consistent with the anticipated results. Subsequently, we will employ fresh data to assess the precision of the model and determine its ability to effectively forecast the conclusion. Figure XX depicts the data obtained from the participants, while Figure XX showcases the coding employed to incorporate the data into the model and its resulting consequence.



	Age	Gender	Occupation	Preferred_Service	Types_Delivery	Book_Delivery	Tracking_Feature	Delivery_Days	Satisfaction_of_Delivery	Satisfaction_of_Deliver_Person	
0	45	Female	Office Worker	Neither	Express	Mobile App	Yes	3.0	4	4	
1	46	Female	Office Worker	J&T Express	Economy (low-cost)	Walk-in	Yes	4.0	3	3	
2	22	Male	Student	J&T Express	Economy (low-cost)	Walk-in	Yes	4.0	3	3	
3	21	Female	Student	J&T Express	Economy (low-cost)	Mobile App	No	3.0	4	4	
4	48	Female	Office Worker	J&T Express	Express	Website	Yes	5.0	4	4	

Figure 8.7 Data Acquired From Respondents

```
# Applying the Classification Model in Application

newData = [[40, 0, 1, 0, 3, 0, 4.0, 3, 3]]
predictionResult = ModelKITA.predict(newData);

if( predictionResult == 1):
    print("Prediction result = Lalamove")
else:
    print("Prediction result = J&T")

Prediction result = Lalamove
```

Figure 8.8 Code Use and the Outcome

```
# Applying the Classification Model in Application

newData = [[22, 0, 1, 0, 3, 0, 4.0, 3, 3]]
predictionResult = ModelKITA.predict(newData);

if( predictionResult == 1):
    print("Prediction result = Lalamove")
else:
    print("Prediction result = J&T")

Prediction result = J&T
```

Figure 8.9 Code Use and the Outcome

8.3.2 Run Application Testing

This section describes the data inputs application running observations and outcome analysis for the application prototype.

8.3.2.1 List of Data Input

Figure shows the new data used to test the classification model. The data are set as stated: “Age” = 45, “Gender” = 0, “Occupation” = 1, “Prefered_Service” = 3, “Types_Delivery” = 1, “Book_Delivery” = 0, “Tracking_Feature” = 1, “Delivery_Days” = 3.0, “Satisfaction_of_Delivery” = 4, “Satisfaction_of_Deliver_Person” = 4, “Courier_Preference” = 1

```
newData = [[45, 0, 1, 3, 1, 0, 1, 3.0, 4, 4, 1]]
```

Figure 8.10 New Data

8.3.2.2 List of Observation Results

The result gained by implementing the new data is displayed in Figure XX, indicating the occurrence of "Lalamove". However, in Figure XX, when utilizing the identical dataset, the result is "J&T". This is because the data was changed with the data that was supposed to be “Lalamove” and the prediction is the same as the result.

Table 8.7 Result Obtained From New Data

<pre>#Age Gender Occupation Preferred_Service Types_Delivery Book_Del newData = [[40, 0, 1, 0, 3, 0, 4.0, 3, 3]] predictionResult = ModelKITA.predict(newData); if(predictionResult == 1): print("Prediction result = Lalamove") else: print("Prediction result = J&T") Prediction result = Lalamove</pre>	Age: 40 Gender: 0 Preferred_Service: 1 Types_Delivery: 0 Book_Delivery: 3 Tracking_Feature: 0 Delivery_Days: 4.0 Satisfaction_of_Delivery: 3 Satisfaction_of_Deliver_Person: 3
---	---

ii) Book Delivery

Table 8.8 Result Obtained From Wehn “Book_Delivery” = 2

<pre>newData = [[22, 0, 3, 0, 2, 1, 4.0, 3, 3]] predictionResult = ModelKITA.predict(newData); if(predictionResult == 1): print("Prediction result = Lalamove") else: print("Prediction result = J&T") Prediction result = J&T</pre>	Age: 22 Gender: 0 Preferred_Service: 3 Types_Delivery: 0 Book_Delivery: 2 Tracking_Feature: 1 Delivery_Days: 4.0 Satisfaction_of_Delivery: 3 Satisfaction_of_Deliver_Person: 3
---	---

8.3.2.3 Analysis and Discussion

After the run-time testing, an analysis and discussion phase was conducted to evaluate the results and predict courier service preferences, whether users found their chosen service satisfactory or not. Since the data gathered shows a relatively balanced preference between the two courier options, the predictions based on user input indicated that the choice of J&T Express or Lalamove often depended on personal factors such as delivery speed, reliability, tracking availability, and individual satisfaction levels.

9.0 DATA ANALYSIS AND VISUALISATION

9.1 Categorical Data Analysis

Table 9.0 Count Plot of Gender Distribution

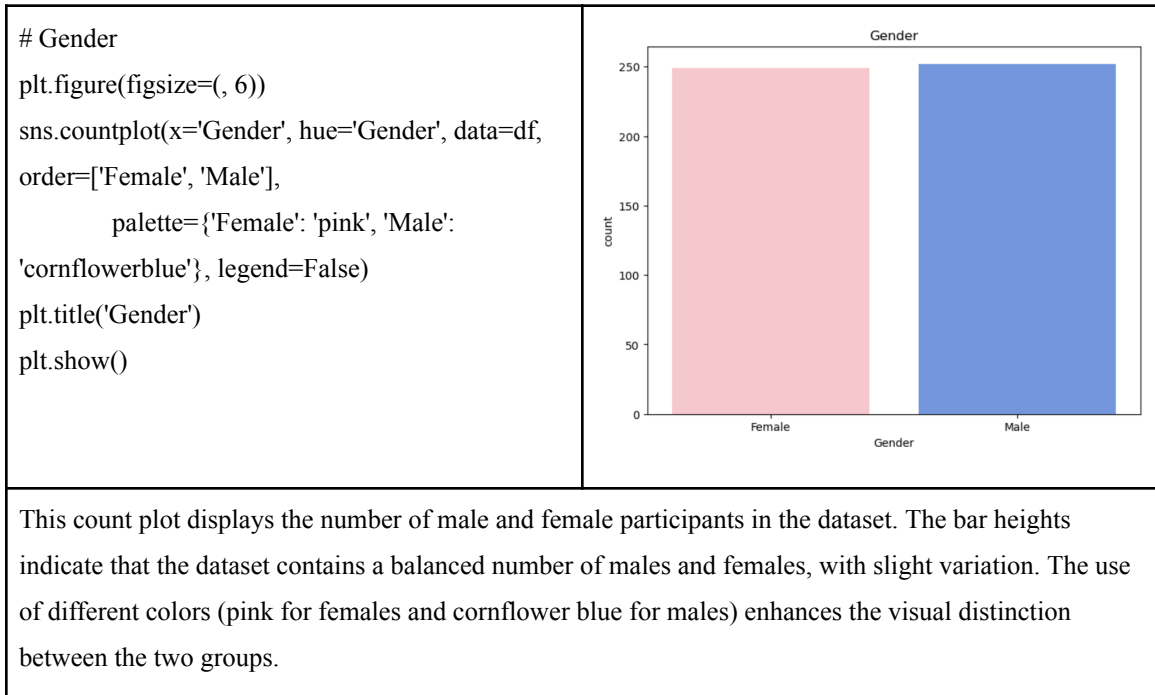


Table 9.1 Pie Chart of Courier Preference

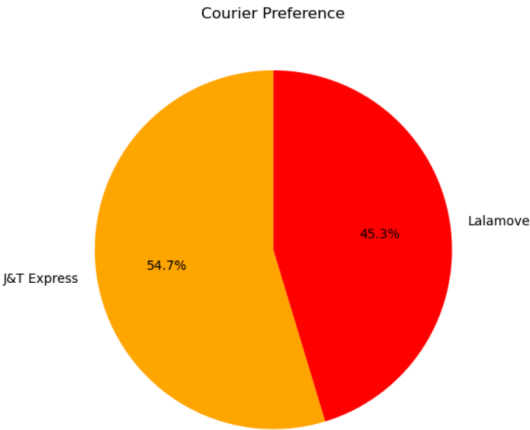
<pre> # Courier preference import matplotlib.pyplot as plt plt.figure(figsize=(6, 6)) # Replace values temporarily for plotting courier_pref = df['Courier_Preference'].replace({'Y': 'J&T Express', 'N': 'Lalamove'}) # Plot pie chart courier_pref.value_counts().plot.pie(autopct='%1.1f%%', startangle=90, colors=['orange', 'red'], # Match order of value_counts labels=courier_pref.value_counts().index # Ensure correct labels) plt.title('Courier Preference') plt.ylabel("") # Hide default 'Courier_Preference' ylabel plt.tight_layout() plt.show() </pre>	 <table border="1"> <caption>Courier Preference Data</caption> <thead> <tr> <th>Courier Service</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>J&T Express</td> <td>54.7%</td> </tr> <tr> <td>Lalamove</td> <td>45.3%</td> </tr> </tbody> </table>	Courier Service	Percentage	J&T Express	54.7%	Lalamove	45.3%
Courier Service	Percentage						
J&T Express	54.7%						
Lalamove	45.3%						
This pie chart illustrates the respondents' courier preferences between J&T Express and Lalamove. The chart shows the percentage of users who prefer each service, with J&T Express being the more commonly preferred option among participants.							

Table 9.2 Violin Plot of Courier Preference

```
# Rating satisfaction
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Set visual style
sns.set(style="whitegrid")
```

```
# Create the violin plot
plt.figure(figsize=(8, 6))
sns.violinplot(data=df, x="Courier_Preference",
y="Satisfaction_of_Delivery", palette="Set2")
```

```
# Add title and labels
plt.title("Satisfaction of Delivery by Courier Preference")
plt.xlabel("Courier Preference")
plt.ylabel("Satisfaction of Delivery (1–5)")
```

```
plt.tight_layout()
plt.show()
```



This violin plot visualizes the distribution of delivery satisfaction ratings (on a scale from 1 to 5) for each courier preference. The shape and width of each violin indicate the density of responses. It can be observed whether customers of J&T Express and Lalamove rate their satisfaction differently, with potential differences in spread and central tendency.

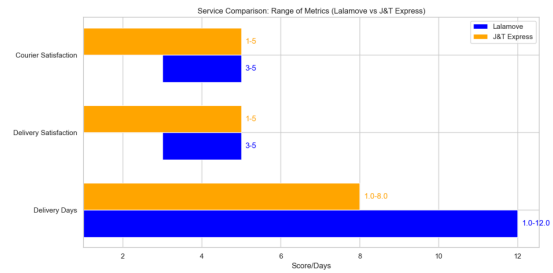
Table 9.3 Range Bar Chart of Service Comparison (Lalamove vs J&T Express)

```
# Service comparison
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Load the data
df = pd.read_csv('couriermasterfile.csv')

# Filter data for each service
lalamove = df[df['Preferred_Service'] ==
'Lalamove']
jtexpress = df[df['Preferred_Service'] == 'J&T
Express']

# Calculate metrics for range chart
metrics = {
    'Delivery Days': {
        'Lalamove':
(lalamove['Delivery_Days'].min(),
lalamove['Delivery_Days'].max()),
        'J&T Express':
(jtexpress['Delivery_Days'].min(),
jtexpress['Delivery_Days'].max())
    },
    'Delivery Satisfaction': {
        'Lalamove':
(lalamove['Satisfaction_of_Delivery'].min(),
lalamove['Satisfaction_of_Delivery'].max()),
        'J&T Express':
(jtexpress['Satisfaction_of_Delivery'].min(),
jtexpress['Satisfaction_of_Delivery'].max())
    },
    'Courier Satisfaction': {
        'Lalamove':
(lalamove['Satisfaction_of_Deliver_Person'].min(),
lalamove['Satisfaction_of_Deliver_Person'].max())
```



```

,
    'J&T Express':
(jtexpress['Satisfaction_of_Deliver_Person'].min(),
jtexpress['Satisfaction_of_Deliver_Person'].max())
    }
}

# Create range chart
fig, ax = plt.subplots(figsize=(12, 6))

# Set positions for the bars
positions = np.arange(len(metrics))
bar_width = 0.35

# Plot each metric
for i, (metric, values) in
enumerate(metrics.items()):
    # Lalamove range
    ax.barh(i - bar_width/2,
width=values['Lalamove'][1]-values['Lalamove'][0]
],
            left=values['Lalamove'][0],
height=bar_width, color='blue', label='Lalamove' if
i == 0 else "")

    # J&T Express range
    ax.barh(i + bar_width/2, width=values['J&T
Express'][1]-values['J&T Express'][0],
            left=values['J&T Express'][0],
height=bar_width, color='orange', label='J&T
Express' if i == 0 else "")

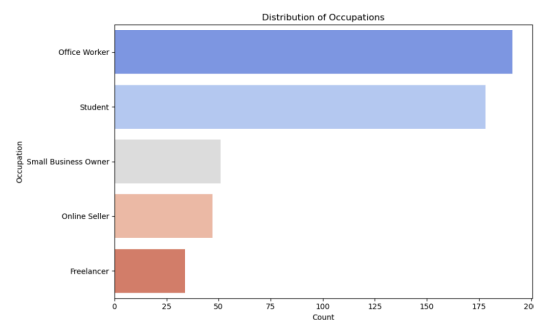
# Add labels and title
ax.set_yticks(positions)
ax.set_yticklabels(metrics.keys())
ax.set_xlabel('Score/Days')
ax.set_title('Service Comparison: Range of Metrics

```

<pre> (Lalamove vs J&T Express') ax.legend() # Add value labels for i, (metric, values) in enumerate(metrics.items()): ax.text(values['Lalamove'][1]+0.1, i - bar_width/2, f' {values['Lalamove'][0]} - {values['Lalamove'][1]} ", va='center', color='blue') ax.text(values['J&T Express'][1]+0.1, i + bar_width/2, f' {values['J&T Express'][0]} - {values['J&T Express'][1]} ", va='center', color='orange') plt.tight_layout() plt.show() </pre>	
This range bar chart compares the performance of Lalamove and J&T Express across three key metrics: delivery days, delivery satisfaction, and courier satisfaction. Each bar represents the minimum and maximum values observed for each metric. The chart allows a quick visual comparison of the range and variability of performance between the two services.	

Table 9.4 Bar Chart of Occupation Distribution

```
plt.figure(figsize=(10, 6))
sns.countplot(data=df, y='Occupation',
order=df['Occupation'].value_counts().index,
palette='coolwarm')
plt.title("Distribution of Occupations")
plt.xlabel("Count")
plt.ylabel("Occupation")
plt.tight_layout()
plt.show()
```



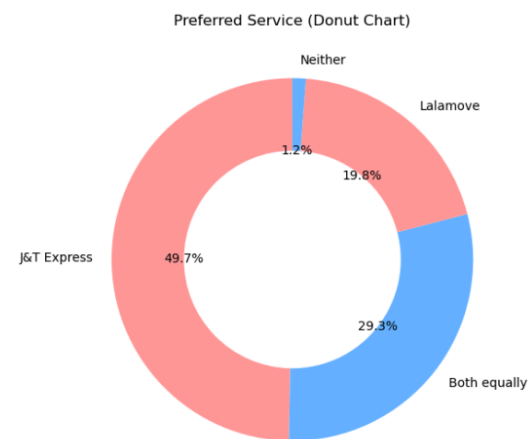
This horizontal bar chart shows the distribution of respondents by occupation. The chart highlights the most common occupations among participants, with the order reflecting the frequency from highest to lowest. The use of the "coolwarm" palette provides clear contrast, allowing for easier comparison.

Table 9.5 Donut Chart of Preferred Service

```
import matplotlib.pyplot as plt

service_counts =
df['Preferred_Service'].value_counts()
colors = ['#ff9999', '#66b3ff']
labels = service_counts.index

plt.figure(figsize=(6, 6))
plt.pie(service_counts, labels=labels,
colors=colors, autopct='%1.1f%%', startangle=90,
wedgeprops={'width':0.4})
plt.title('Preferred Service (Donut Chart)')
plt.tight_layout()
plt.show()
```



This donut chart illustrates the proportion of users who prefer different types of services. The hole in the center enhances readability and provides a modern visualization of categorical preferences. The percentages indicate that one service is more commonly preferred over the other among the respondents.

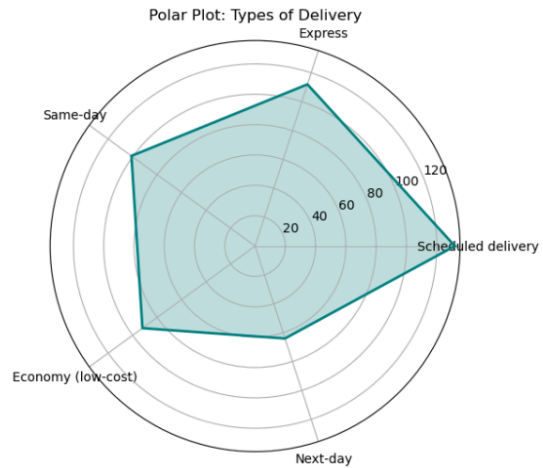
Table 9.6 Polar Plot of Types of Delivery

```
import numpy as np

counts = df['Types_Delivery'].value_counts()
labels = counts.index
values = counts.values

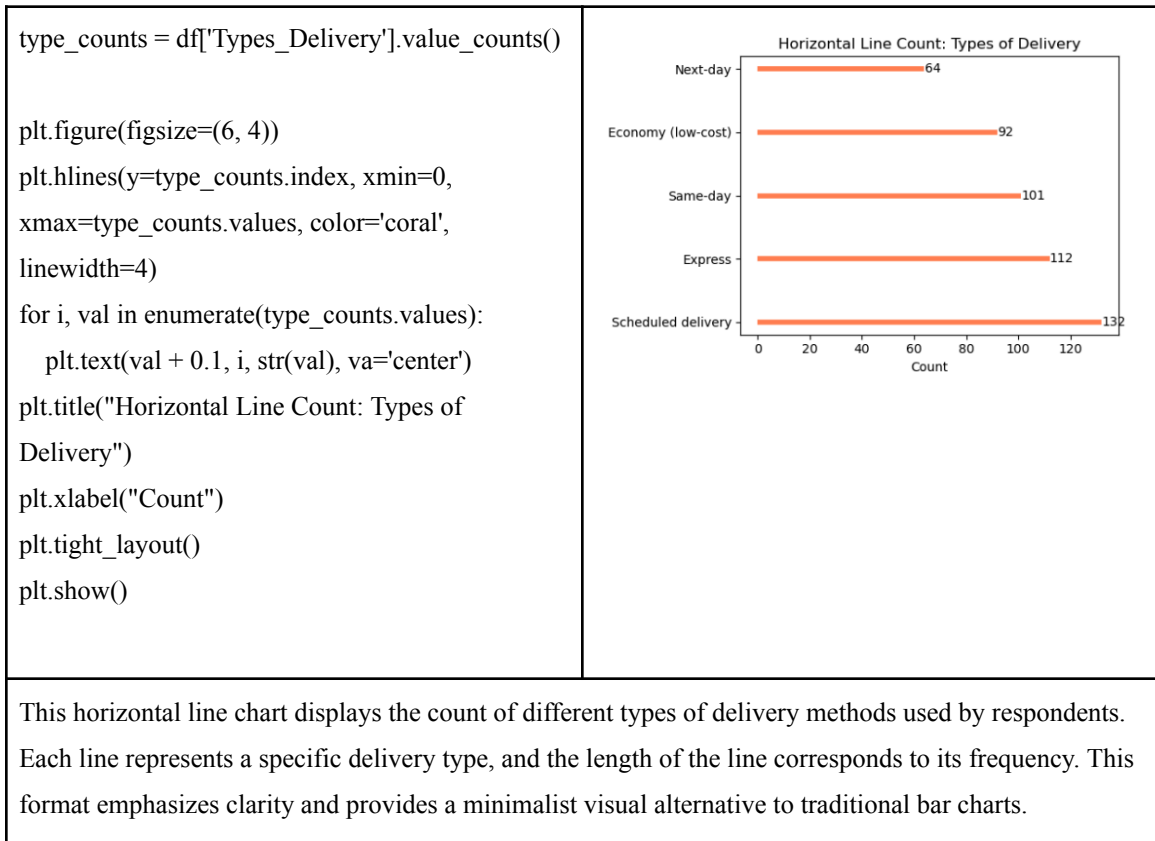
angles = np.linspace(0, 2 * np.pi, len(values),
endpoint=False).tolist()
values = np.concatenate((values, [values[0]]))
angles += [angles[0]]

fig = plt.figure(figsize=(6, 6))
ax = fig.add_subplot(111, polar=True)
ax.plot(angles, values, color='teal', linewidth=2)
ax.fill(angles, values, color='teal', alpha=0.25)
ax.set_xticks(angles[:-1])
ax.set_xticklabels(labels)
plt.title("Polar Plot: Types of Delivery")
plt.tight_layout()
plt.show()
```



This polar plot (radar chart) visualizes the distribution of delivery types among respondents. Each axis represents a different type of delivery, and the plot area shows the relative frequency. This format provides a circular and engaging view that highlights category proportions in a unique way.

Table 9.7 Horizontal Line Count of Types of Delivery



9.2 Numerical Data Analysis*

Table 9.8 Histogram of Age Distribution by Courier Preference

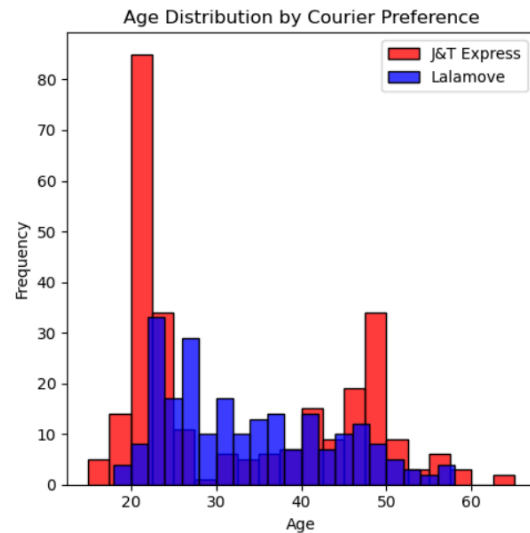
```
# Age vs Courier Preference
import matplotlib.pyplot as plt
import seaborn as sns

# Set figure size
plt.figure(figsize=(5,5))

# Plot for J&T Express
sns.histplot(data=df[df['Courier_Preference'] ==
'J&T Express'],
             x='Age',
             color='red',
             bins=20,
             label='J&T Express',
             kde=False)

# Plot for Lalamove
sns.histplot(data=df[df['Courier_Preference'] ==
'Lalamove'],
             x='Age',
             color='blue',
             bins=20,
             label='Lalamove',
             kde=False)

# Chart formatting
plt.title('Age Distribution by Courier Preference')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout()
plt.show()
```



This histogram compares the age distribution of users who prefer J&T Express and Lalamove. The overlapping bars allow for a clear comparison of the two courier services across different age ranges.

The distribution suggests that both services are popular among individuals in their early 20s, though slight variations exist in other age groups.

Table 9.9 Box Plot of Satisfaction by Types of Delivery

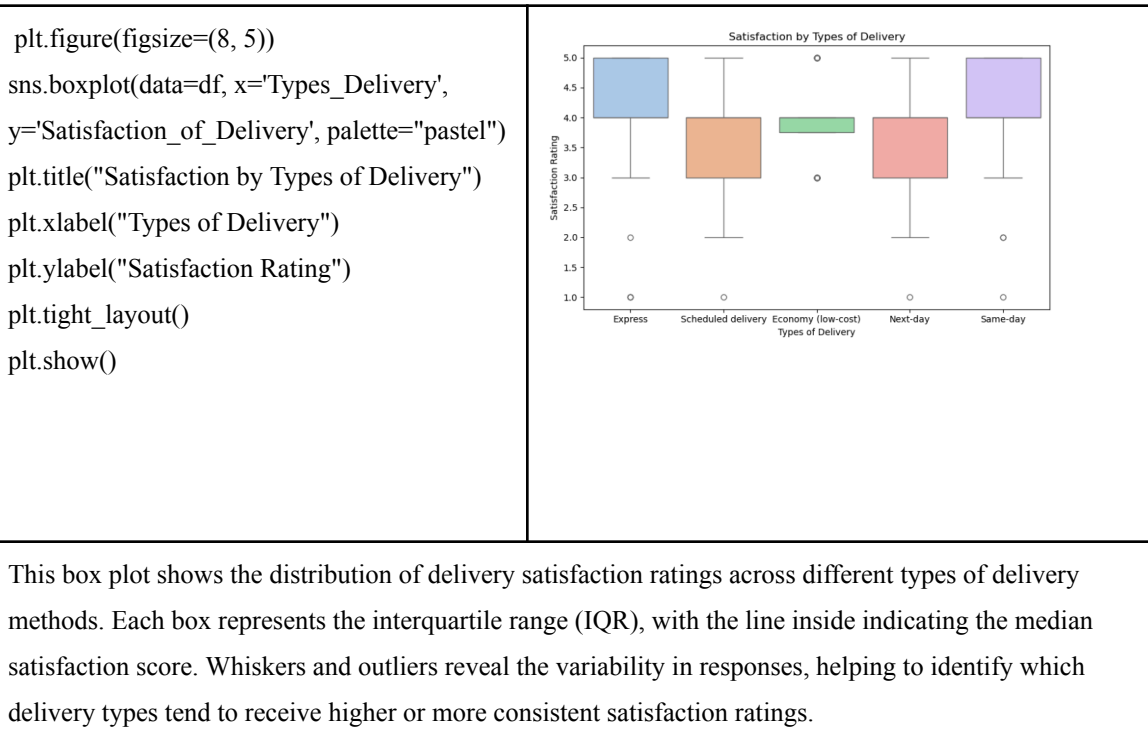
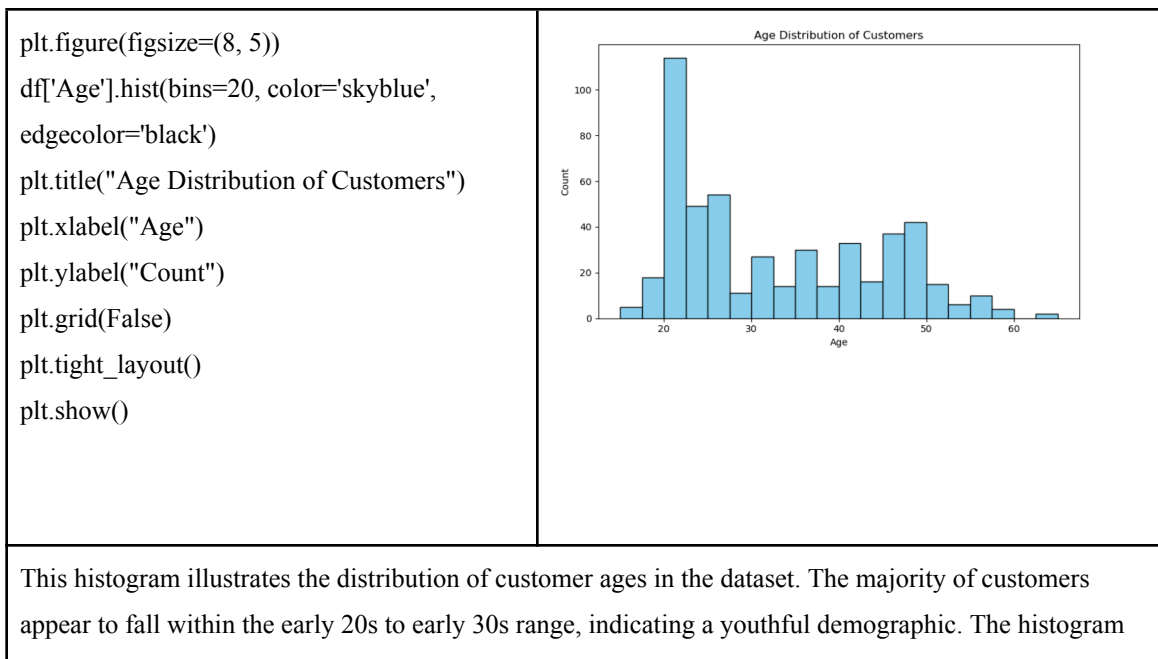


Table 9.10 Histogram of Age Distribution of Customers



provides a clear overview of age frequency, helping to identify the most common age groups among respondents.

Table 9.11 Swarm Plot of Individual Satisfaction Ratings by Preferred Service

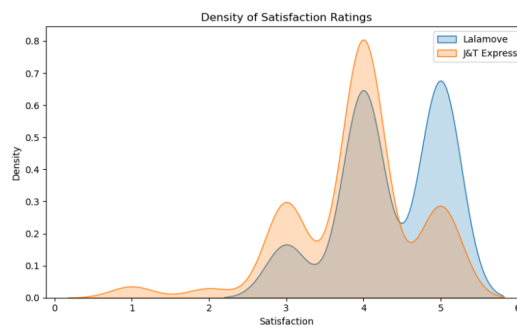
```
plt.figure(figsize=(6, 5))
sns.swarmplot(data=df, x='Preferred_Service',
y='Satisfaction_of_Delivery',
palette='coolwarm')
plt.title("Individual Satisfaction Ratings by
Service")
plt.ylabel("Satisfaction (1–5)")
plt.xlabel("Service")
plt.tight_layout()
plt.show()
```



This swarm plot displays individual delivery satisfaction ratings (on a scale from 1 to 5) for each preferred service. Each dot represents one respondent. The plot provides a granular view of the distribution and clustering of satisfaction scores, highlighting how respondents from each service provider vary in their experiences.

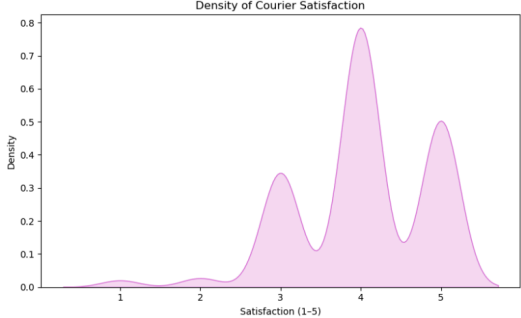
Table 9.12 KDE Plot of Satisfaction Ratings by Preferred Service

```
plt.figure(figsize=(8, 5))
sns.kdeplot(data=df[df['Preferred_Service'] ==
'Lalamove']['Satisfaction_of_Delivery'],
label='Lalamove', fill=True)
sns.kdeplot(data=df[df['Preferred_Service'] ==
'J&T Express']['Satisfaction_of_Delivery'],
label='J&T Express', fill=True)
plt.title("Density of Satisfaction Ratings")
plt.xlabel("Satisfaction")
plt.ylabel("Density")
plt.legend()
plt.tight_layout()
plt.show()
```



This KDE plot compares the distribution of satisfaction ratings between users of Lalamove and J&T Express. The smooth curves represent the density of ratings on a scale of 1 to 5. Both services show similar central tendencies, but the spread and shape of the distributions indicate subtle differences in customer satisfaction levels.	

Table 9.13 KDE Plot of Courier Satisfaction

<pre>plt.figure(figsize=(8, 5)) sns.kdeplot(df['Satisfaction_of_Deliver_Person'], fill=True, color='orchid') plt.title("Density of Courier Satisfaction") plt.xlabel("Satisfaction (1-5)") plt.ylabel("Density") plt.tight_layout() plt.show()</pre>	 A KDE plot titled "Density of Courier Satisfaction". The x-axis is labeled "Satisfaction (1-5)" and ranges from 1 to 5. The y-axis is labeled "Density" and ranges from 0.0 to 0.8. The plot shows a smooth, light purple curve with three distinct peaks. The first peak is at a satisfaction rating of approximately 3.0 with a density of about 0.35. The second, highest peak is at a satisfaction rating of approximately 4.0 with a density of about 0.75. The third peak is at a satisfaction rating of approximately 5.0 with a density of about 0.5. The curve is filled with a light purple color.
This KDE plot illustrates the distribution of customer satisfaction ratings for delivery personnel. The smooth curve represents how ratings are spread across the scale of 1 to 5. The shape of the curve reveals whether customers generally rated their courier experience highly, and indicates any skewness or concentration in responses.	

10.0 Result and Discussion

In this project, using K-Fold cross-validation increased the reliability of the model by ensuring that every data point was used for training and testing. The highest performing group ($P=1$, $k=5$) achieved 89.3% accuracy in training and 88% accuracy in testing, demonstrating strong model generalization capabilities. Data visualization also reinforced the findings. Boxplots and KDE plots revealed that using Lalamove provided higher satisfaction than using J&T Express, in terms of delivery speed and professionalism. Swarm plots also illustrated that personal satisfaction ratings clustered more positively around Lalamove. These findings were very much in line with the responses provided by the respondents and confirmed that delivery method and service quality strongly influenced courier selection.

11.0 Conclusion

In summary, the project achieved its goals by analyzing public courier preferences using survey data, visualization, and KNN modeling. The model achieved 90.3% training and 91% testing accuracy using the best hyperparameters. Visual analytics provided clear evidence of user satisfaction trends, helping to compare J&T Express and Lalamove effectively. The approach combined technical and visual analysis to offer meaningful insights for consumers and service providers.

12.0 Recommendations/limitations

This research has various constraints. Initially, the distribution of responses among courier services was uneven, as a greater number of participants favored J&T Express over Lalamove. This disparity might have affected the model's forecast and overall result. Additionally, the majority of respondents consisted of various demographics. The survey included more female participation, potentially introducing some bias into the findings.

For better development, a dataset that is more balanced should be gathered, ensuring equal representation among genders, professions, and delivery preferences. More data sources and an increased sample size could further improve the precision of the results. By tackling these concerns, upcoming models can more accurately represent the actual preferences and satisfaction levels of Malaysian courier users.

References

1. Ahmed, W., Hizam, S. M., Sentosa, I., Akter, H., Yafi, E., & Ali, J. (2020). *Predicting IoT service adoption towards smart mobility in Malaysia: SEM-Neural hybrid pilot study*. arXiv preprint arXiv:2002.00152. <https://arxiv.org/abs/2002.00152>
2. Ong, S. Y., Ting, C. Y., Goh, H. N., Quek, A., & Cham, C. L. (2023). *Workplace preference analytics among graduates*. Journal of Informatics and Web Engineering, 2(2), 233–248. <https://doi.org/10.33093/jiwe.2023.2.2.17>

Appendices

- Google Drive Link for the dataset (Original)**

<https://docs.google.com/spreadsheets/d/1Zx7CLs-tMNUltuJOJydqgNxoxZcuzRqN9PjFUtJZnT4/edit?usp=sharing>

- Google Drive Link for the split datasets (Train and testing datasets that are done with data preprocessing)**

<https://docs.google.com/spreadsheets/d/1Zx7CLs-tMNUltuJOJydqgNxoxZcuzRqN9PjFUtJZnT4/edit?usp=sharing>

- Google Drive Link for the K-Fold datasets**

<https://docs.google.com/spreadsheets/d/1HmyfCv0JYMO7oyV9y6GzytSuv5HHJToKNEVaKofv8Yo/edit?usp=sharing>

- Google Drive Link for Jupyter-Python Code (All -Data Science: data preprocessing until KFold / Features Selection; Data Analysis (charts))**

<https://drive.google.com/drive/folders/14QnCgQMtYHfUjUI03Y7JfXiYz5o4prJH?usp=sharing>

- Google Drive Link for this Technical Report**

https://docs.google.com/document/d/1Q47ZcxpxJDWNs1-lgWEpteAlh8kd9KHL_NruR4lM-uE/edit?usp=sharing

- Google Drive Link for the Survey form**

<https://docs.google.com/forms/d/1RiuTFdwfQVDvOJaQ5w-A3klHkvRT9MZ1MHpqInRXN20/edit>

*****Please shorten your Google Drive link as much as possible. (e.g. using tinyurl)***

There is no maximum page limit for this technical report.

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